

Title: *Securing the Data-Driven Enterprise: A Study on Cybersecurity Challenges in Big Data Frameworks*

Abstract

While Big Data and ML drive innovation, they also expand the attack surface for cyber threats. This paper examines the intersection of IT innovation and data security, focusing on how machine learning can be both a tool for defense and a target for adversarial attacks in large-scale data platforms.

1. The Security Paradox of Big Data

Big data platforms (Hadoop, NoSQL) often prioritize availability and scalability over granular security. As organizations move toward data-driven decision-making, the integrity of the data becomes the most critical asset. If the "Big Data" is poisoned, the "Machine Learning" outcome is fundamentally flawed.

2. Adversarial Machine Learning

This section discusses the rise of adversarial attacks where subtle perturbations in input data can lead an ML model to make incorrect classifications. We review case studies where IT systems in the financial sector were compromised via "model inversion" attacks.

3. Machine Learning for Network Defense

Despite the risks, ML offers the best defense mechanism for modern IT infrastructure.

- **Anomaly Detection:** *Using Isolation Forests to detect zero-day exploits in network traffic.*
- **Automated Response:** *Developing scripts that automatically isolate compromised nodes in a distributed Big Data cluster without shutting down the entire system.*

4. Policy and Governance

Technical solutions must be paired with robust data governance. We analyze the role of "Explainable AI" (XAI) in helping IT managers understand why a security flag was raised, bridging the gap between automated systems and human oversight.

5. Conclusion

A truly innovative IT system is one that is resilient. Integrating security protocols directly into the Big Data lifecycle is no longer optional but a foundational requirement for 2026 and beyond.